Sub-pixel Extrema Refinement and Edge Rejection in SIFT

1 The one-dimensional skeleton

Expand f around a sample point a, and write the offset as $\delta = x - a$:

$$f(a+\delta) \;\approx\; f(a) + f'(a)\,\delta + \tfrac{1}{2} f''(a)\,\delta^2.$$

To locate the critical point, differentiate *with respect to* δ and set the result to zero:

$$\frac{d}{d\delta}\Big[\, f(a) + f'(a)\,\delta + \tfrac{1}{2} f''(a)\,\delta^2 \,\Big] = f'(a) + f''(a)\,\delta = 0.$$

$$\delta = -\frac{f'(a)}{f''(a)}.$$

The refined location is $a + \delta$. This is exactly one Newton step applied to f', which is what setting a quadratic's derivative to zero amounts to.

2 Lifting to three dimensions

Now the coordinate is $\mathbf{x} = (s, y, x)$ and $f = D$ is the DoG value. The single derivative becomes a **gradient** (a 3-vector); the second derivative becomes a **Hessian** (a 3×3 matrix):

$$\mathbf{g} = \nabla D = \begin{bmatrix} D_s \\ D_y \\ D_x \end{bmatrix}, \qquad H = \begin{bmatrix} D_{ss} & D_{sy} & D_{sx} \\ D_{sy} & D_{yy} & D_{yx} \\ D_{sx} & D_{yx} & D_{xx} \end{bmatrix}.$$

With offset vector $\boldsymbol{\delta} = (\delta_s, \delta_y, \delta_x)$, the expansion is

$$D(\mathbf{x}_0 + \boldsymbol{\delta}) \;\approx\; D(\mathbf{x}_0) + \mathbf{g}^\top \boldsymbol{\delta} + \tfrac{1}{2}\,\boldsymbol{\delta}^\top H \boldsymbol{\delta}.$$

This is the 1D formula with each piece promoted: $f'(a)\,\delta \to \mathbf{g}^\top\boldsymbol{\delta}$ (a dot product), and $\frac{1}{2} f''(a)\,\delta^2 \to \frac{1}{2}\boldsymbol{\delta}^\top H \boldsymbol{\delta}$ (a quadratic form).

3 Solving for the offset

Take the gradient with respect to $\boldsymbol{\delta}$ and set it to $\mathbf{0}$. Two standard identities do the work — the multivariable analogues of $\frac{d}{d\delta}(c\,\delta) = c$ and $\frac{d}{d\delta}(\frac{1}{2} q\,\delta^2) = q\,\delta$:

$$\frac{\partial}{\partial \boldsymbol{\delta}}\big(\mathbf{g}^\top \boldsymbol{\delta}\big) = \mathbf{g}, \qquad \frac{\partial}{\partial \boldsymbol{\delta}}\big(\tfrac{1}{2}\boldsymbol{\delta}^\top H \boldsymbol{\delta}\big) = H\boldsymbol{\delta} \quad (H \text{ symmetric}).$$

Therefore

$$\frac{\partial D}{\partial \boldsymbol{\delta}} = \mathbf{g} + H\boldsymbol{\delta} = \mathbf{0},$$

$$H\boldsymbol{\delta} = -\mathbf{g} \qquad \Longrightarrow \qquad \boldsymbol{\delta} = -H^{-1}\mathbf{g}.$$

Compare the 1D box, $\delta = -f'/f''$: identical structure. Here $-H^{-1}$ *is* "divide by the second derivative," except that in three dimensions "divide" means "solve a 3×3 linear system."

4 What you do with the offset

1. Refined location

$$\hat{\mathbf{x}} = \mathbf{x}_0 + \boldsymbol{\delta}.$$

Because s is one of the three axes, the scale is refined too — σ becomes continuous instead of snapping to the discrete level ladder.

2. Refined value at the peak

Substitute $\boldsymbol{\delta} = -H^{-1}\mathbf{g}$ back into the expansion. The quadratic term simplifies using $H\boldsymbol{\delta} = -\mathbf{g}$:

$$\boldsymbol{\delta}^\top H\boldsymbol{\delta} = \boldsymbol{\delta}^\top (H\boldsymbol{\delta}) = \boldsymbol{\delta}^\top(-\mathbf{g}) = -\,\mathbf{g}^\top\boldsymbol{\delta}.$$

Hence

$$D(\hat{\mathbf{x}}) = D + \mathbf{g}^\top\boldsymbol{\delta} + \tfrac{1}{2}\boldsymbol{\delta}^\top H\boldsymbol{\delta} = D + \mathbf{g}^\top\boldsymbol{\delta} - \tfrac{1}{2}\mathbf{g}^\top\boldsymbol{\delta},$$

$$D(\hat{\mathbf{x}}) = D(\mathbf{x}_0) + \tfrac{1}{2}\,\mathbf{g}^\top\boldsymbol{\delta}.$$

The peak of a parabola sits half a gradient-step above the sample — the same $\frac{1}{2}$ as in 1D. Threshold $|D(\hat{\mathbf{x}})| < 0.03$ to reject low-contrast points.

3. Convergence / re-centering

If $\max_i |\delta_i| > 0.5$, the true peak is nearer a different sample. Move $\mathbf{x}_0$ to that integer neighbour, recompute $\mathbf{g}$ and H there, and re-solve. Iterate at most 5 times; discard points that fail to converge or wander outside the volume. Every derivative is thus always evaluated at an integer grid point — only the final residual offset is fractional.

5 Finite-difference stencils

The gradient and Hessian are filled in by central differences on the integer grid (spacing 1):

$$D_x = \frac{D[x{+}1] - D[x{-}1]}{2}, \qquad D_{xx} = D[x{+}1] - 2D[x] + D[x{-}1],$$

$$D_{xy} = \frac{D[y{+}1, x{+}1] - D[y{+}1, x{-}1] - D[y{-}1, x{+}1] + D[y{-}1, x{-}1]}{4}$$

(indices along the other axes suppressed). The mixed-partial stencil touches all corners of the $3 \times 3 \times 3$ cube — which is why the border voxels must be dropped before this step.

6 Edge rejection via the Hessian

DoG responds strongly all along an edge, producing extrema strung out down a ridge that slide under small viewpoint changes. The discriminator is curvature. Take the *spatial* 2×2 Hessian (drop the scale row and column):

$$H_{2\times 2} = \begin{bmatrix} D_{xx} & D_{xy} \\ D_{xy} & D_{yy} \end{bmatrix}.$$

Its eigenvalues α, β are the principal curvatures. A ridge has one large and one small; a blob has both large. Their ratio $r = \alpha/\beta$ can be tested without an eigendecomposition, using only trace and determinant:

$$\frac{\mathrm{Tr}(H_{2\times 2})^2}{\det(H_{2\times 2})} = \frac{(\alpha+\beta)^2}{\alpha\beta} = \frac{(r+1)^2}{r}.$$

This depends only on the ratio, not on the individual magnitudes, and is minimised at $r = 1$. Keep a point when

$$\frac{\mathrm{Tr}(H_{2\times 2})^2}{\det(H_{2\times 2})} < \frac{(r+1)^2}{r}, \qquad r = 10 \quad \left(\text{threshold } = \tfrac{121}{10} = 12.1\right).$$

Reject $\det(H_{2\times 2}) \le 0$ outright: opposite-sign curvatures mean a saddle, not an extremum, and guard against dividing by near-zero.

Pipeline order

Refine first, then test contrast on the refined value $D(\hat{\mathbf{x}})$, then apply edge rejection. Doing edge rejection before refinement tests the Hessian at the wrong location.